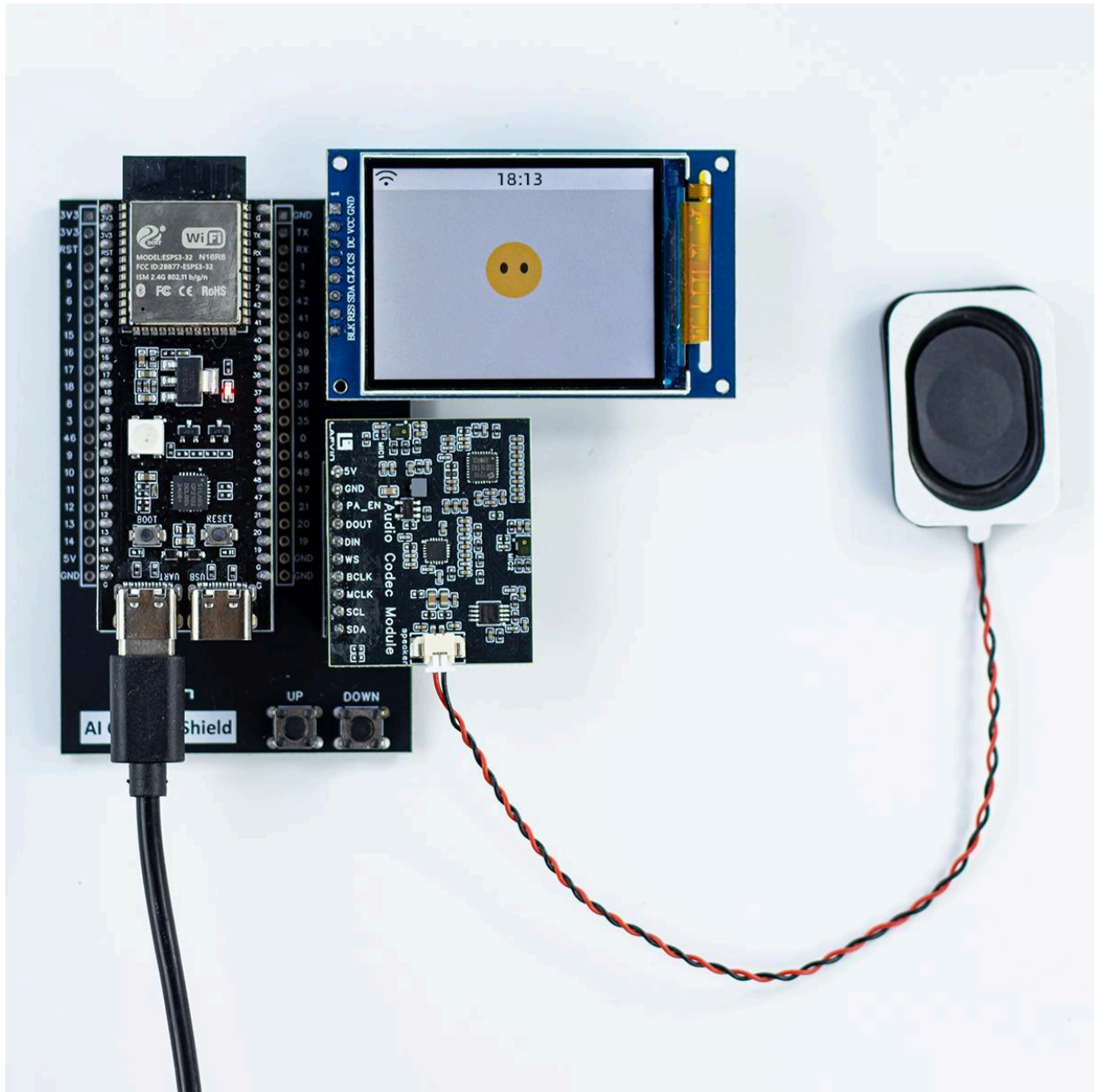
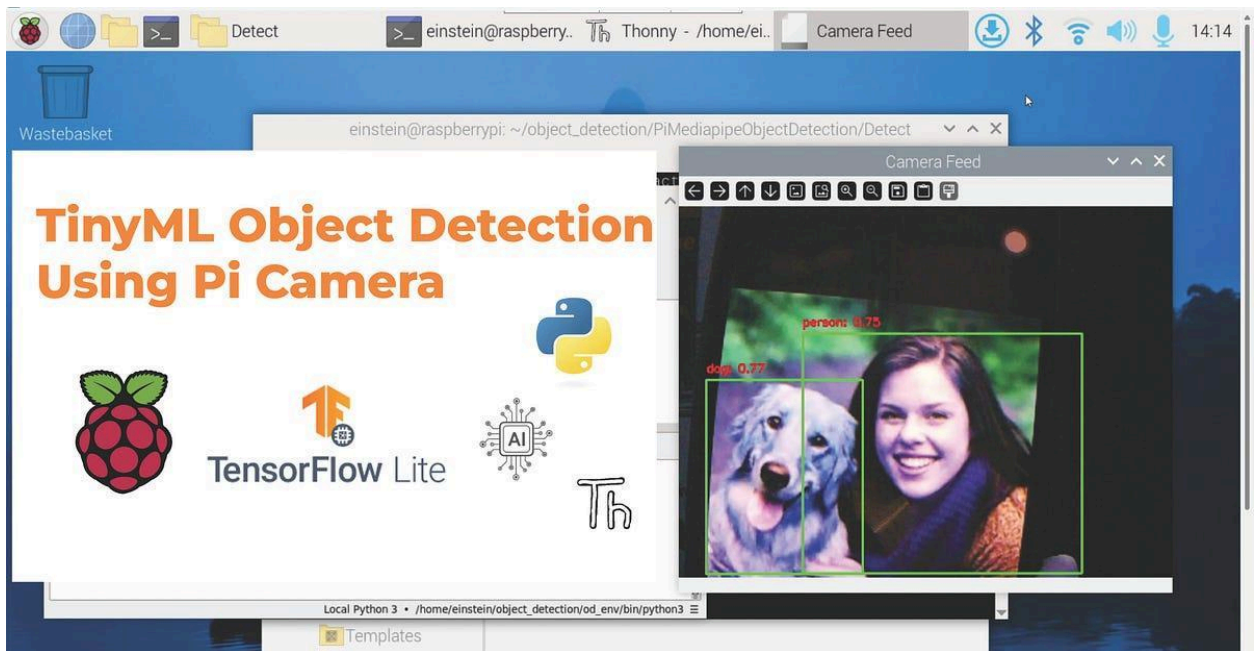
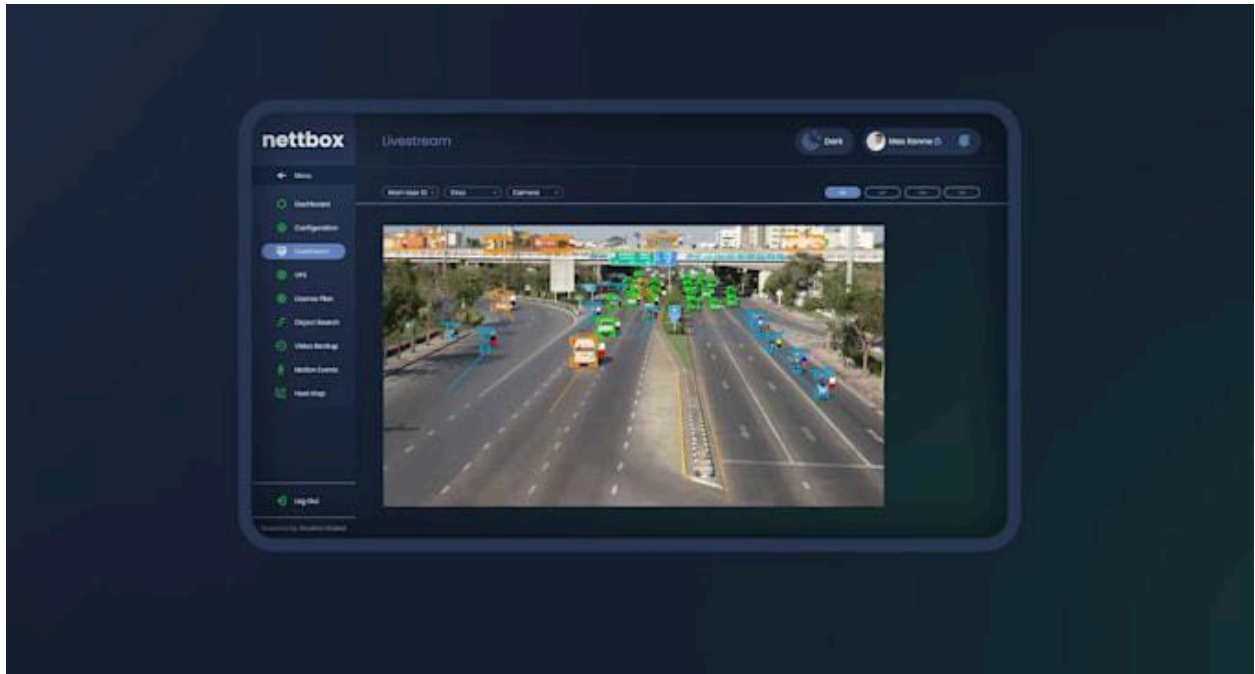


## High-Demand Embedded AI Skills (2026)











# Real Video Analytics



## 1. OpenCV

**Purpose:** Computer Vision

Learn:

- Image Processing
- Video Processing
- Motion Detection
- Object Tracking

- Face Detection

Projects:

- Smart CCTV
- Vehicle Counter
- Thermal Monitoring

`pip install opencv-python`

---

## 2. YOLO

**Purpose:** Real-Time Object Detection

Learn:

- Object Detection
- Segmentation
- Tracking
- Custom Training

Projects:

- Human Detection
- Drone Surveillance
- Traffic Monitoring

`pip install ultralytics`

---

## 3. Deep Learning

Learn:

- Neural Networks
- CNN
- RNN
- Transfer Learning

Frameworks:

- TensorFlow

- PyTorch

Projects:

- Face Recognition
  - Medical AI
  - Industrial Inspection
- 

## 4. Edge AI ⚡

Run AI locally without cloud.

Devices:

- Raspberry Pi
- Jetson
- ESP32

Benefits:

- Faster Response
- Offline Operation
- Lower Cost

Projects:

- Smart Camera
  - AI Robot
  - Security System
- 

## 5. Embedded Linux 🐧

Learn:

- Linux Commands
- Shell Scripting
- GPIO Control
- Device Drivers
- Networking

Practice On:

- Raspberry Pi Raspberry Pi 5

Useful Commands:

ls  
cd  
pwd  
top  
htop  
systemctl

---

## 6. ESP32

Learn:

- GPIO
- WiFi
- Bluetooth
- MQTT
- Sensor Integration

Projects:

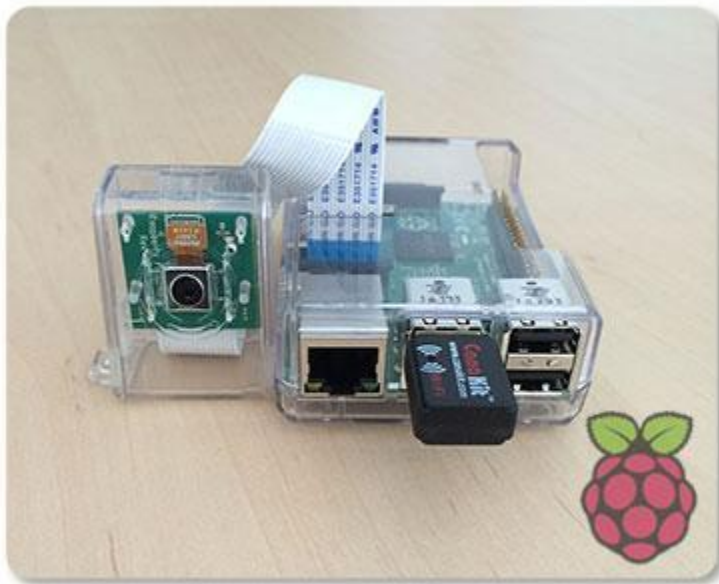
- Smart Agriculture
- AI Door Lock
- Environmental Monitoring

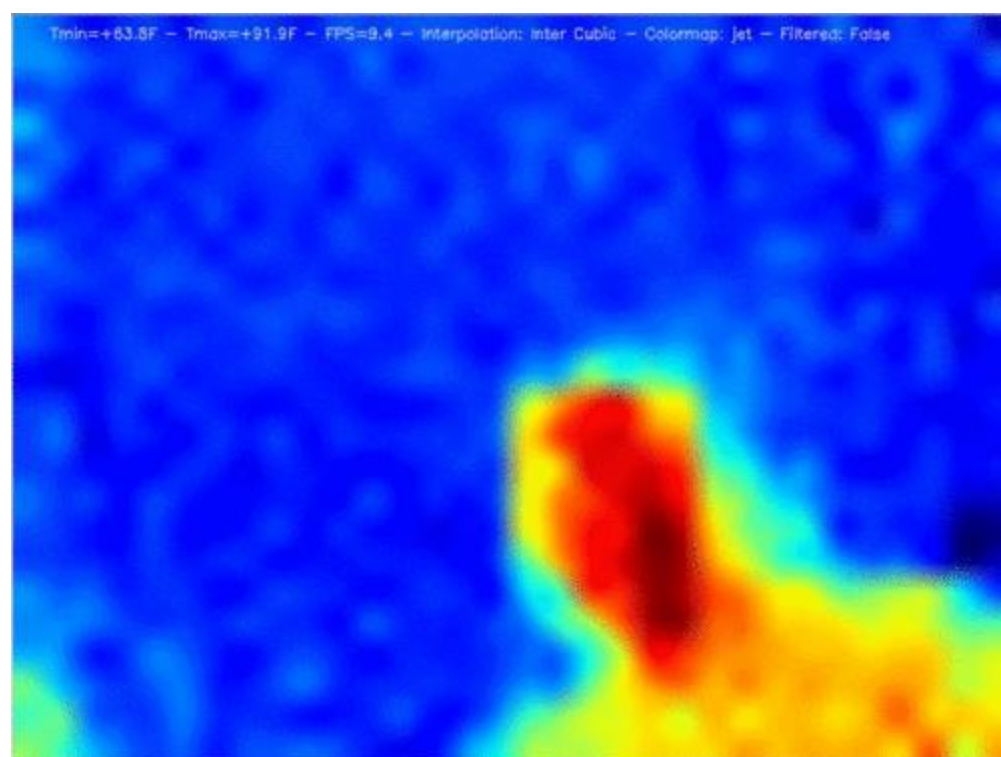
Tools:

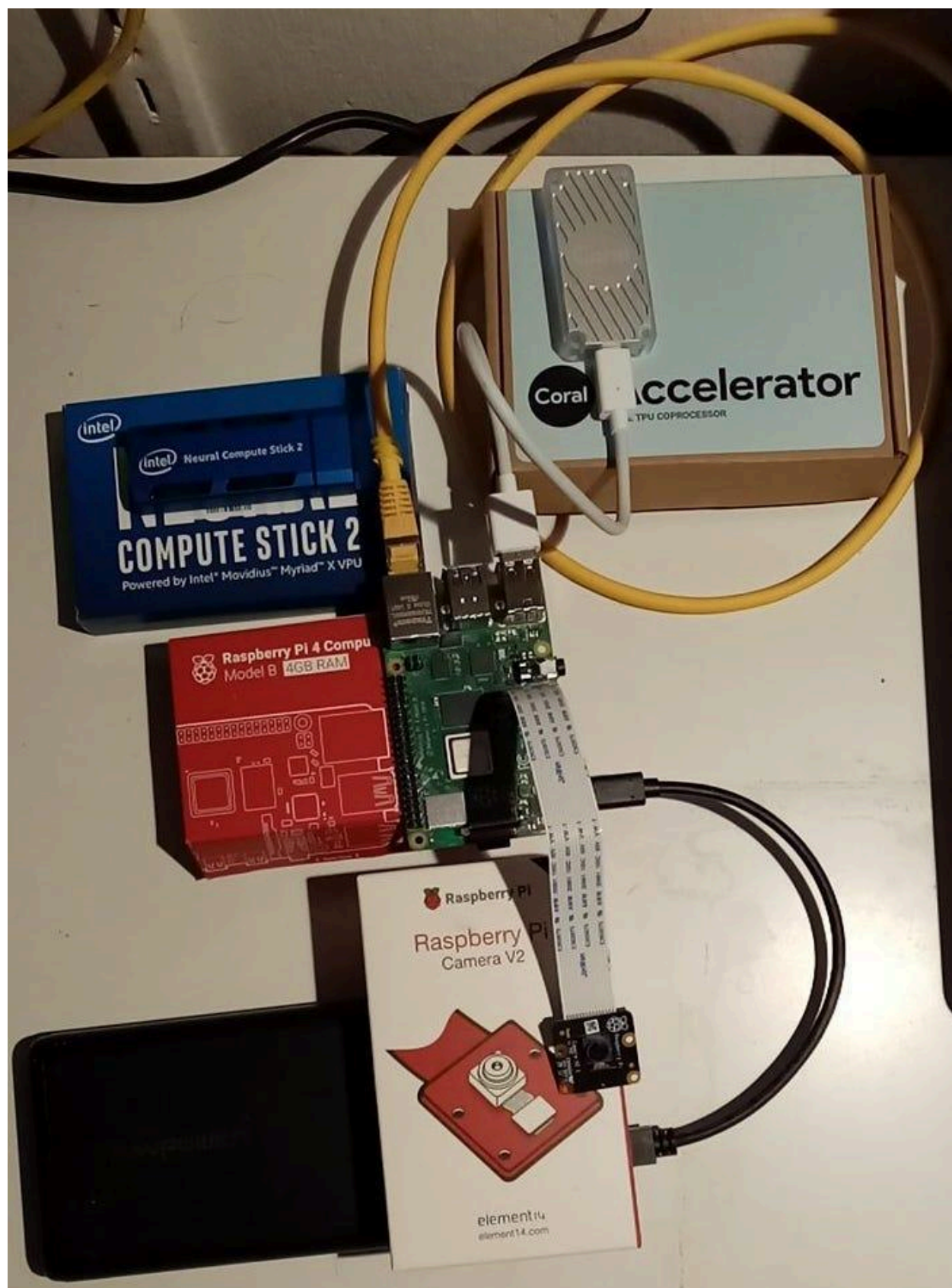
- Arduino IDE
  - ESP-IDF
- 

## 7. Raspberry Pi









Learn:

- Linux
- Python
- Camera Programming
- I2C
- SPI
- UART

Projects:

- AI CCTV
- Thermal Camera System
- LoRa Gateway

---

## 8. TensorFlow Lite

Purpose:

Run AI models on edge devices.

Learn:

- Model Conversion
- Quantization
- Optimization

Projects:

- Mobile AI
- Embedded AI
- TinyML

`pip install tensorflow`

Convert:

```
converter = tf.lite.TFLiteConverter.from_saved_model("model")
```

---

## 9. Real-Time Video Analytics

Learn:

- Video Streams
- Object Tracking
- Event Detection
- Alerts

Projects:

- Smart Surveillance
- Crowd Monitoring
- Traffic Analytics

Tech Stack:

- OpenCV
  - YOLO
  - DeepSORT
- 

## 10. MLOps

Learn:

- Git
- Docker
- FastAPI
- CI/CD
- Kubernetes Basics

Projects:

- AI Deployment Server
- Model Monitoring
- Cloud AI APIs

`pip install fastapi uvicorn`

---

## 11. Generative AI + Vision Models

Learn:

- LLMs
- Vision-Language Models
- Multimodal AI
- AI Agents

Popular Models:

- OpenAI GPT
- Google Gemini
- Meta Llama

Projects:

- AI Assistant
  - Visual Question Answering
  - Smart Surveillance Reports
- 

## 12. TinyML

Run AI on microcontrollers.

Devices:

- ESP32
- STM32
- Arduino

Learn:

- Quantization
- TensorFlow Lite Micro
- Edge Inference

Projects:

- Gesture Recognition
  - Voice Recognition
  - Predictive Maintenance
-